

THE STRUCTURE OF MATHEMATICAL REALITY

Phase IV — Chapter 11

Probability, Universality, and Levels of Description

Quiz

Self-Study Curriculum

Before you begin

This quiz tests application, not memorisation. Answer in your own words.

Do not quote the lecture back at yourself. If you find yourself writing a phrase you remember reading, stop and rephrase it from your own understanding. A short, precise answer in your own language is worth more than a long transcription.

1. A colleague says: “Probability is just long-run frequency, the Bayesian stuff is philosophy.” Give the strongest case for why a single-patient prediction forces you past the frequency interpretation. Then give the strongest case *for* the frequentist.

2. Two very different biological systems are well described by the same simple equation. From this chapter, give the *least* mystical explanation for why. What has that explanation ruled out about how much you understand either system?

3. You are told that averaging a certain measurement across samples is meaningless. Give two structurally different reasons this could be true, one from *this* chapter (concentration) and one you expect from the *next* (measurement scale).

4. In what precise sense is universality a *partial* answer to the unreasonable-effectiveness problem of Chapter 6? What does it leave completely untouched?

After the quiz

For each answer, ask yourself: “*Did I explain the **why**, or did I just state the **what**?*” If you stated a fact without explanation, go back and add the reasoning.

Date: _____ **Questions where I want to go deeper:**
